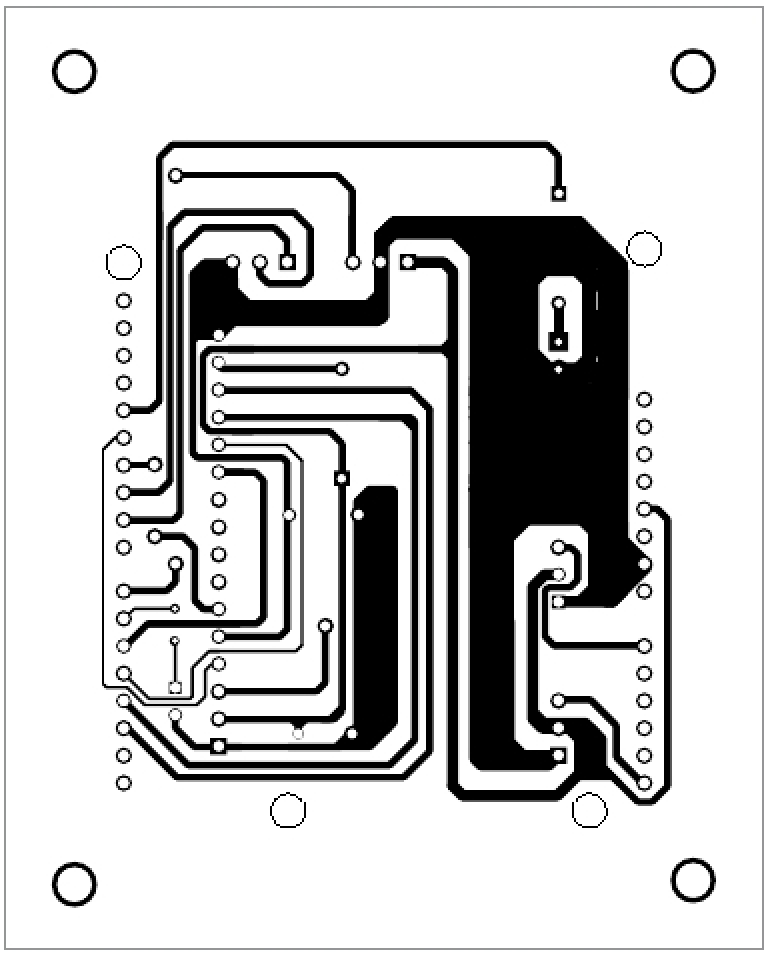

Fig. 2: Actual-size PCB layout of multi-utility protection system

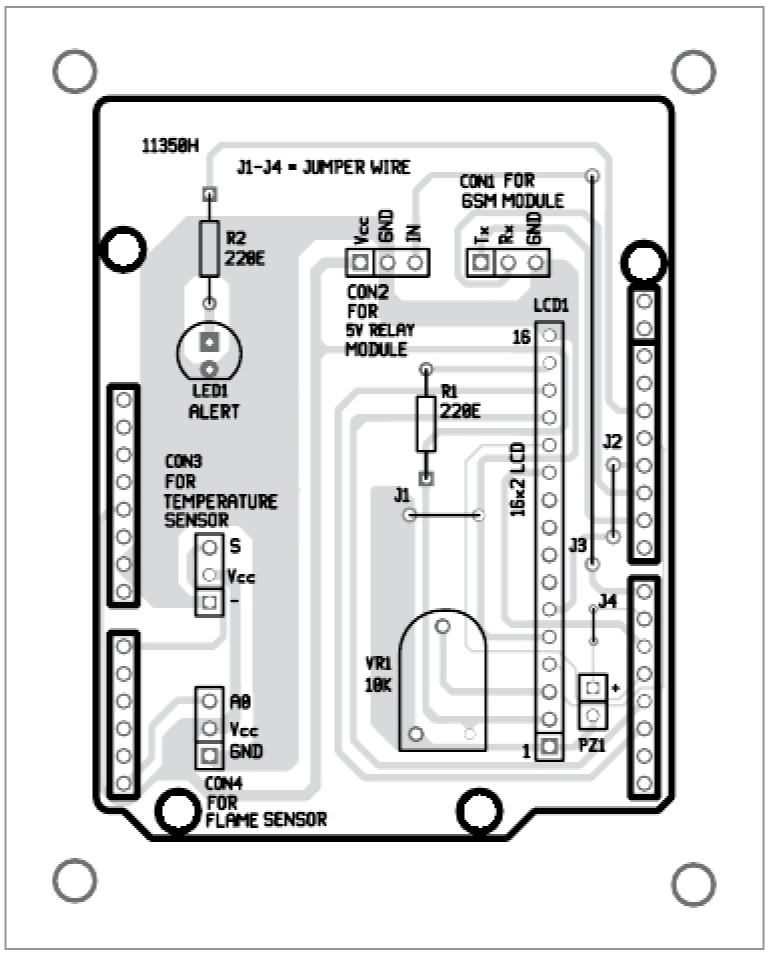

Fig. 3: Components layout for the PCB

of 9600 to 115200. It contains internal TCP/IP stack to enable connection with the Internet via GPRS. It is suitable for SMSes, calls and Internet connectivity applications.

Temperature sensor. NTC thermistor-based KY-013 sensor module is used here. It is an analogue temperature sensor module compatible with Arduino. A thermistor is a low-cost, small device. It is sensitive to ambient temperature variation. It is used to detect the temperature of the surrounding environment. KY-013 can be directly connected to analogue pin A0 of Arduino to detect temperature changes in the environment. The temperature measuring range is between 20°C and 80°C.

Preset. A preset is a three-terminal resistor with a rotating contact that makes an adjustable voltage divider. If only two terminals are utilised—one end and the wiper—it behaves like a variable resistor or rheostat. A 10-kilo-ohm preset (VR1) is used here.

Software

Programming of the project is done on Arduino IDE version 1.8.6. Pin A5 of Arduino for the flame sensor and pin A0 of Arduino for the temperature sensor are defined in Arduino program code/sketch (agriculture.ino). The program displays flame and temperature levels on the serial monitor in the PC and on the LCD in digital form.

Construction and testing

An actual-size PCB layout of the multi-utility protection system is shown in Fig. 2 and its components layout in Fig. 3. Assemble the components on the PCB. Make connections as per the circuit diagram.

Note that 32-pin male Berg connectors should be connected on the PCB from solder side and should be soldered on the components side. Then, insert this board into the 32-pin female Berg connectors on top of the Arduino board.

Switch on the power supply by connecting 9V battery to the circuit. Check flame level on the LCD and the serial monitor. If there is a burning flame or sparks in motors, transformers or other devices, the circuit will send a buzzer alert, update the status of flame and temperature levels on the LCD, cut off the load by de-energing the relay, and the system will send an SMS to the registered cellphone.

For initial testing, light a candle and bring its flame near the flame sensor. Monitor the flame detection distance against the flame values shown on the serial monitor. Sensitivity can be adjusted using inbuilt potentiometer in KY-026 module.

On pressing S on the serial monitor, the system starts recording the flame intensity level. If the value goes beyond the threshold level, the buzzer sounds an alarm and LED1 starts blinking. At the same time, it sends an SMS alert to the cell number included in the software code. The flame level can also be viewed on the serial monitor and the LCD.

Test the circuit in various situations and make sure it is working well before use. EFY

EFY Note

The source code of this project is available for free download at *source.efymag.com*

PARTS LIST

Semiconductors:		
Board1	-	Arduino Uno board
LCD1	-	16x2 LCD (JHD162A)
GSM1	-	SIM900A GSM module
LED1	-	5mm LED
KY-013	-	3-pin temperature sensor module
KY-026	-	3-pin flame sensor module
Resistors (all 1/4-watt, ±5% carbon):		
R1, R2	-	220-ohm
VR1	-	10-kilo-ohm preset
Miscellaneous:		
BATT.1	-	9V battery
PZ1	-	Piezo buzzer
CON1-CON4	-	3-pin female Berg connector
RL1	-	5V relay module
	-	Jumper wires
	-	Berg strips
	-	32-pin male Berg connector
	-	16-pin female Berg connector

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